Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 1 (Generated practice paper - Set 34)

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

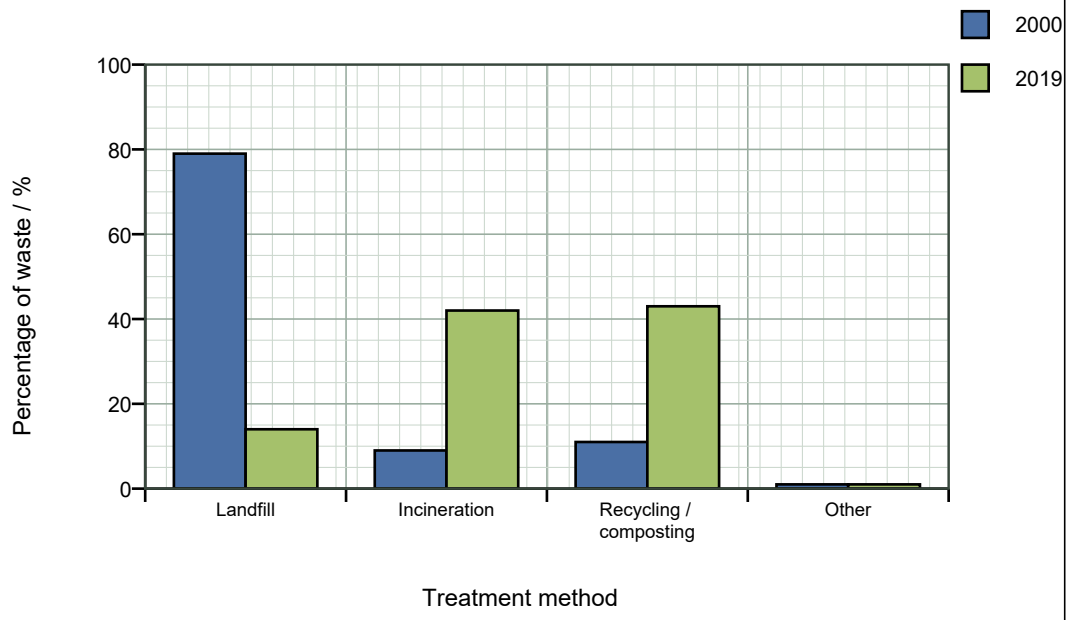
Question	Mark
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11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 Figure 1 shows how municipal solid waste was treated in the UK in 2000 and 2019.

Figure 1



0 1 . 1 Describe the changes shown in Figure 1.

[2 marks]

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0 1 . 2 Explain **three** environmental disadvantages of landfill that led to the change.

[3 marks]

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0 1 . 3 Describe **four** features of a well-managed landfill site.

[4 marks]

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0 1 . 4 Suggest why incineration has increased more than recycling for some waste types.

[1 mark]

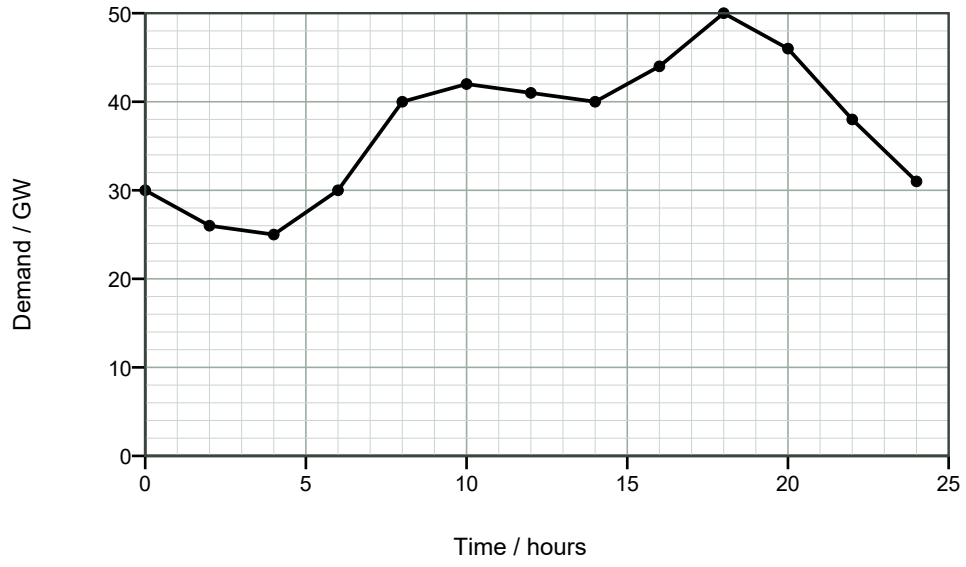
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0 2 Figure 1 shows the demand for electricity in the UK over a 24-hour period on a weekday in winter.

Figure 1



0 2 . 1 Describe the pattern of demand shown in Figure 1.

[2 marks]

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0 2 . 2 Explain the causes of the fluctuations shown in Figure 1 and give **two** other causes of fluctuations in electricity demand.

[4 marks]

0 2 . 3 Explain why base-load power stations (nuclear, large coal) produce a surplus of electricity at night.

[2 marks]

0 2 . 4 Give **two** causes of fluctuations in energy **supply**.

[2 marks]

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0 3 . 1 Explain how the control of emission **location** and emission **timing** can reduce the severity of pollution. Use examples.

[6 marks]

Emission location

Emission timing

0 3 . 2 Explain the **polluter pays** principle and suggest why it encourages pollution prevention.

[2 marks]

0 3 . 3 Explain the **precautionary principle** as applied to the release of a new chemical.

[2 marks]

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0 4 Table 1 shows information about two copper deposits, X and Y, being assessed for mining. The current cut-off ore grade for copper is 0.4%.

Table 1

Feature	Deposit X	Deposit Y
Ore grade / % copper	0.9	0.35
Depth of deposit / m	350	40
Overburden	Hard igneous rock	Loose glacial sediment
Distance to nearest railway / km	120	8
Annual rainfall / mm	2 100	500
Chemical form	Copper sulfide	Copper sulfide

0 4 . 1 Explain what is meant by the **cut-off ore grade** and state which deposit is currently included in the copper reserves.

[2 marks]

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0 4 . 2 Use Table 1 to discuss the factors that affect the viability of mining each deposit.

[6 marks]

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0 4 . 3 State **two** changes that would decrease the cut-off ore grade.

[2 marks]

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0 5 A student estimated the population of ground beetles in a field using the Lincoln Index. 60 beetles were caught in pitfall traps, marked with a spot of paint and released. Three days later 45 beetles were caught, of which 9 were marked.

0 5 . 1 Calculate the estimated population size.

[2 marks]

Answer beetles

0 5 . 2 State **four** assumptions of the Lincoln Index method.

[4 marks]

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0 5 . 3 Explain how the estimate would be affected if the paint made the beetles more likely to be eaten by birds.

[2 marks]

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0 5 . 4 Suggest why the second sample was taken after three days rather than the next day or after a month.

[2 marks]

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0 6 Table 1 shows features of oil tanker design and operation that reduce the risk of oil pollution.

0 6 . 1 Complete Table 1 by explaining how each feature reduces oil pollution.

[5 marks]

Table 1

Feature	How it reduces oil pollution	
Double hull		
Inert gas system		
Separate oil and ballast tanks		
Twin engines / rudders		
Recirculation of oil during unloading (crude oil washing)		

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0 6 . 2 Explain how each of the following reduces the risk of tanker accidents.

[2 marks]

Offshore shipping routes .....

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GPS / Automatic Identification System .....

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0 6 . 3 Describe how a **bund wall** prevents oil pollution from a storage tank.

[2 marks]

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0 6 . 4 Explain the purpose of an oil interceptor on the drainage system of a car park.

[1 mark]

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0 7 Table 1 shows the water quality required for four different uses.

0 7 . 1 Complete Table 1 by giving the quality requirement for each water use and the reason for it.

[4 marks]

Table 1

Water use	Quality requirement	Reason	
Potable (public) supply			
Spray irrigation			
Power station condenser cooling			
Industrial boiler water			

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0 7 . 2 Explain why aeration is used in the treatment of water from anaerobic sources.

[2 marks]

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0 7 . 3 Explain how flocculation removes clay particles that sedimentation cannot.

[2 marks]

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0 7 . 4 Give **two** methods used to sterilise treated water.

[2 marks]

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0 8 . 1 Which soil property is most directly responsible for capillary rise of water from below? Tick one box.

[1 mark]

A High sand content ☐

B High clay content ☐

C Crumb peds ☐

D High pH ☐

0 8 . 2 Distinguish between hygroscopic, capillary and gravitational water in soil and state which are available to plants.

[3 marks]

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0 8 . 3 Explain why crumb peds produce more fertile soil than platy peds.

[1 mark]

5

0 9 . 1 Complete Table 1 to show one method of controlling each source of railway noise.

[4 marks]

Table 1

Source of railway noise	Control method	
Wheel vibration on the track		
Pantograph turbulence		
Wheel squeal on corners		
Braking squeal		

0 9 . 2 Explain why road traffic noise affects more people than any other form of noise pollution.

[1 mark]

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0 9 . 3 Describe **three** ways of reducing road traffic noise reaching residents.

[3 marks]

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0 9 . 4 Suggest how noise from pile-driving at sea for wind turbine foundations can be reduced to protect cetaceans.

[2 marks]

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1 0 . 1 Evaluate the extent to which new technologies could allow fossil fuels and nuclear fission to continue to supply a large proportion of the world's energy sustainably.

[9 marks]

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1 0 . 2 State the approximate percentage of natural gas in a reservoir that can be recovered using its own pressure.

[1 mark]

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Write an essay on one of the following topics.

1 1 . 1 Discuss the difficulties in monitoring and predicting global climate change and evaluate the methods used to overcome them.

[25 marks]

OR

1 1 . 2 Discuss how knowledge of the properties of energy resources can be used to reduce the severity of their impact on the environment.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS